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Nationality: Chinese.

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Research interests: low-carbon cementitious materials, thermodynamic modelling, reactive/diffusive transport, cement durability, UHPC/UHDC self-healing, engineered barriers for radioactive-waste disposal, machine learning, multi-objective optimisation, X-ray CT, computational materials science.

Strengths: GEMS thermodynamic modelling, FEniCSx/FEM, Python scientific computing, HPC, X-ray CT image analysis, machine learning, experimental data analysis, diffusion/transport modelling.

DEGREES

11.2020-04.2024	Doctor of Philosophy (Ph.D.), Structural, Seismic and Geotechnical Engineering. Supervisors: Prof. Liberato Ferrara; Prof. Marco di Prisco Politecnico di Milano, Milan, Italy. Research focus: <i>Self-healing and durability of ultra-high durability / ultra-high performance concrete under aggressive environments and sustained loading.</i>
09.2017-06.2020	Master of Science (M.Sc.), Civil Engineering. Supervisor: Prof. Yingwu Zhou Shenzhen University, Shenzhen, China.
09.2013-06.2017	Bachelor of Engineering (B.Eng.), Civil Engineering. Yunnan Agricultural University, Kunming, China.

WORK EXPERIENCE

07.2024-present	Paul Scherrer Institute (PSI), Villigen, Switzerland. Postdoctoral Researcher. Laboratory for Waste Management (LES). Supervisor: Dr. Nikolaos I. Prasianakis Research on thermodynamic modelling, cement–bentonite–host-rock interfaces, diffusion and reactive transport, XCT-based 3D modelling, FEniCSx, GEMS, HPC, and AI-assisted low-carbon cement design.
09.2025	Hiroshima University, Japan. Visiting Scholar. Research exchange and academic collaboration in cementitious materials and infrastructure engineering.
11.2020-04.2024	Politecnico di Milano, Milan, Italy. Ph.D. Researcher. Supervisor: Prof. Liberato Ferrara Research on UHPC/UHDC self-healing, durability, aggressive exposure, sustained loading, and data-driven modelling within the EU Horizon 2020 project ReSHEALience.
06.2020-10.2020	Shenzhen University, Shenzhen, China. Research Assistant. Supervisor: Prof. Yingwu Zhou Research on engineered cementitious composites (ECC) and sustainable low-carbon cementitious materials.
06.2018-06.2019	Shenzhen University, Shenzhen, China. Teaching Assistant. Teaching Assistant for the course Concrete Structures at Shenzhen University.

RESEARCH PROJECTS

- EURAD-2 (European Partnership on Radioactive Waste Management), Work Package SUDOKU, Near-Surface Disposal Optimization based on Knowledge and Understanding, 2024-present, ongoing, Researcher / Task contributor. Degradation, HTO diffusion experiments, XCT-based pore-scale models and reactive-transport modelling of cementitious barrier systems.
- EURAD-2 (European Partnership on Radioactive Waste Management), Work Package OPTI, Optimisation of Repository Support Structures, 2025-present, ongoing, Researcher / Task contributor. Concrete mix optimisation for tunnel linings using machine-learning models of carbonation and strength with NSGA-II multi-objective optimisation.
- SCENE, Swiss Center of Excellence on Net Zero Emissions, 2024-2026, ongoing, Researcher. Simulation-driven low-carbon cement design integrating GEMS, machine learning, carbon accounting and multi-objective optimisation.

- EU Horizon 2020, ReSHEALience, Ultra High Durability Concrete, 2020-2024, Completed, Ph.D. Researcher. UHDC/UHPC durability, self-healing, mechanical recovery and performance under severe environments.
- China–Italy Joint Scholarship Program, 2020-2024, Completed, Scholarship recipient / Researcher.

PROFESSIONAL ACTIVITIES

• Editorial

- ✧ **Editorial Board Member:** Journal of the Mechanical Behavior of Materials (JMBM), De Gruyter Brill.
- ✧ **Young Editorial Board Member:** ZKG International – Cement, Lime, Gypsum.
- ✧ **Guest editor**
 - ◆ RIMNI: *Machine Learning-Driven Net-Zero Design and Performance Optimization of Sustainable Cementitious Materials*.
 - ◆ Buildings: *Machine Learning-Driven Modeling and Optimization in Structural Engineering*.
 - ◆ Urban Science: *Practices in Urban Circular Economy Transition*.

• Reviewers for

- ✧ Cement and Concrete Research, Cement and Concrete Composites, Construction and Building Materials, Materials and Structures, Journal of Materials in Civil Engineering (ASCE), KSCE Journal of Civil Engineering, Journal of Cleaner Production, etc.

PERSONAL SKILLS & CERTIFICATES

- Languages: Chinese (mother tongue), English (professional working proficiency), Italian (beginner).
- Main Software: Computing: Python (scikit-learn), Linux, HPC workflows | Thermodynamic modelling: GEMS | Numerical simulation: FEniCSx.
- Numerical / materials modelling: thermodynamic equilibrium modelling, diffusion and reactive transport, inverse modelling.
- Characterization / data: X-ray CT, 3D image segmentation, diffusion experiments, parameter fitting, scientific data analysis.

SELECTED AWARDS & HONOURS

2019	National Scholarship for Graduate Students	Ministry of Education of China
2019	First-Class Graduate Scholarship	Shenzhen University
2018	First-Class Graduate Scholarship	Shenzhen University
2017	Special-Class Scholarship	Shenzhen University
2017	Outstanding Undergraduate Thesis Award	Yunnan Agricultural University
2017	Outstanding Graduate Award	Yunnan Agricultural University

JOURNAL PUBLICATIONS (*Corresponding author)

- [1] W. Huang, J. He, Y. Jin, H. Chen, K. Zhou, **B. Xi**, F. Zhu, D. Lei, DIC and X-CT investigation of damage characterization and microstructural degradation in the concrete interfacial transition zone after elevated temperature exposure. *Structures* 92 (2026) 112771.
- [2] **B. Xi**, N. Zhang, J. He, et al., Thermodynamics-informed C–S–H gel densification descriptor for concrete creep prediction: A physics-informed hybrid framework. *Construction and Building Materials* 538 (2026) 147048.
- [3] M. Davolio, **B. Xi**, L. Ferrara, Tensile constitutive law identification for ultra-high-performance concrete: interchangeability of test methods. *Materials and Structures* 59 (2026) 40.
- [4] J. He, Z. Gao, Y. Xiang, J. Chen, G. Yang, **B. Xi**, D. Lei, H. Zhao, Microscale fracture analysis of concrete interfacial transition zone based on digital image correlation. *Engineering Fracture Mechanics* (2026) 112000.
- [5] J. He, Z. Gao, H. Yin, G. Yang, Y. Xiang, **B. Xi**, D. Lei, H. Zhao, Fluorescent digital image correlation for characterizing interfacial transition zone in fly ash-modified concrete. *Journal of Building Engineering* (2026) 116319.
- [6] N. Zhang, H. Li, **B. Xi**, J. Fan, H. Duan, The environmental burden of building on the Moon. *Nature Astronomy* 9 (2025) 1412–1414.
- [7] Y. Qiu, E. Li, P. Segarra, **B. Xi**, J. Zhou, Developing hybrid XGBoost model to predict the strength of polypropylene and straw fibers reinforced cemented paste backfill and interpretability insights. *Computer Modeling in Engineering & Sciences* (2025).
- [8] M.H. Riaz, Y. Zhou, M. Guo, **B. Xi**, S. Xiang, Development of LC3-ECC incorporating recycled fine aggregate: a novel sustainable and ductile material. *Case Studies in Construction Materials* (2025).
- [9] L. Pang, **B. Xi**, X. Wei, D. Wang, Utilizing electric furnace ferronickel slag as a supplementary cementitious material: Hydration and Cr leaching risk. *Construction and Building Materials* (2025).

- [10] R. Boiger, **B. Xi**, G.-D. Miron, M. Bonvin, J.L. Provis, S.V. Churakov, N.I. Prasianakis, Machine learning-accelerated discovery of green cement recipes. *Materials and Structures* (2025).
- [11] N. Zhang, G. Schiller, H. Azarijafari, Z. Cao, **B. Xi**, et al., Contribution of concrete nitrogenation to global NO_x uptake. *Nature Cities* (2024).
- [12] **B. Xi**, L. Ferrara, Evolution of self-healing performance of UHPC exposed to aggressive environments and cracking/healing cycles. *Materials and Structures* 57 (2024) 36.
- [13] E. Li, N. Zhang, **B. Xi***, Z. Yu, Y. Fissaha, et al., Analysis and modelling of gas relative permeability in reservoir by hybrid KELM methods. *Earth Science Informatics* 17 (2024) 3163–3190.
- [14] **B. Xi**, Z. Huang, S. Al-Obaidi, L. Ferrara, Healing capacity of Ultra High Performance Concrete under sustained load through crack tensile stresses and aggressive environments. *Cement and Concrete Composites* (2023).
- [15] **B. Xi**, S. Al-Obaidi, L. Ferrara, Effect of different environments on the self-healing performance of Ultra High-Performance Concrete – A systematic literature review. *Construction and Building Materials* (2023).
- [16] **B. Xi**, Z. Huang, S. Al-Obaidi, L. Ferrara, Predicting ultra high-performance concrete self-healing performance using hybrid models based on metaheuristic optimization techniques. *Construction and Building Materials* (2023).
- [17] **B. Xi**, N. Zhang, H. Duan, et al., Optimization of rice husk ash concrete design towards economic and environmental assessment. *Environmental Impact Assessment Review* (2023).
- [18] **B. Xi**, E. Li, Y. Fissaha, J. Zhou, et al., LGBM-based modelling scenarios for compressive strength of recycled aggregate concrete with SHAP analysis. *Mechanics of Advanced Materials and Structures* (2023).
- [19] **B. Xi**, N. Zhang, et al., A comprehensive comparison of different regression techniques and nature-inspired optimization algorithms to predict carbonation depth of recycled aggregate concrete. *Frontiers of Structural and Civil Engineering* (2023).
- [20] **B. Xi**, J. He, H. Li, Integration of machine learning models and metaheuristic algorithms for predicting compressive strength of waste granite powder concrete. *Materials Today Communications* (2023).
- [21] E. Li, N. Zhang, **B. Xi***, V.W.Y. Tam, J. Wang, J. Zhou, Intelligent prediction of engineered cementitious composites with limestone calcined clay cement (LC3-ECC) compressive strength based on novel machine learning techniques. *Computers and Concrete* (2023).
- [22] E. Li, N. Zhang, **B. Xi***, J. Zhou, X. Gao, Compressive strength prediction and optimization design of sustainable concrete based on squirrel search algorithm-extreme gradient boosting technique. *Frontiers of Structural and Civil Engineering* (2023).
- [23] Y. Zhou, G. Gong, **B. Xi**, M. Guo, F. Xing, C. Chen, Sustainable lightweight engineered cementitious composites using limestone calcined clay cement (LC3). *Composites Part B: Engineering* (2023).
- [24] J. He, D. Lei, Z. She, **B. Xi**, Flexural performance and damage evaluation on basalt fiber reinforced polymer (BFRP) sheet reinforced concrete. *Construction and Building Materials* (2023).
- [25] J. He, D. Lei, Z. She, **B. Xi**, Investigation on bonding behavior of basalt fiber reinforced polymer (BFRP) sheet reinforced concrete beam. *Journal of Building Engineering* (2023).
- [26] E. Li, **B. Xi**, N. Zhang, X. Shi, J. Zhou, P. Segarra, H. Wang, Circular use of fine-grained tailings to underground mine wind walls. *Circular Economy* (2023).
- [27] N. Zhang, **B. Xi**, J. Li, L. Liu, G. Song, Utilization of CO₂ into recycled construction materials: A systematic literature review. *Journal of Material Cycles and Waste Management* (2022).
- [28] Y. Zhou, S. Zheng, X. Huang, **B. Xi**, Z. Huang, M. Guo, Performance enhancement of green high-ductility engineered cementitious composites by nano-silica incorporation. *Construction and Building Materials* (2021).
- [29] **B. Xi**, Y. Zhou, K. Yu, L. Sui, B. Hu, F. Xing, Use of nano-SiO₂ to develop a high performance green lightweight engineering cementitious composite containing fly ash cenospheres. *Journal of Cleaner Production* (2020).
- [30] Y. Zhou, **B. Xi**, L. Sui, S. Zhen, F. Xing, L. Li, Development of high strain-hardening lightweight engineered cementitious composites: Design and performance. *Cement and Concrete Composites* (2019).
- [31] Y. Zhou, **B. Xi**, K. Yu, L. Sui, F. Xing, Mechanical properties of hybrid ultra-high performance engineered cementitious composites incorporating steel and polyethylene fibers. *Materials* (2018).

CONFERENCE PROCEEDINGS

- [1] **B. Xi**, et al., Machine learning-assisted low-carbon cement design: A path to lower CO₂ emissions and enhanced carbon sequestration, ETH Net Zero Day, ETH Zurich, Zurich, Switzerland, 7 May 2026.
- [2] **B. Xi**, et al., Machine learning assisted low-carbon cement design: A path to lower CO₂ emission and enhanced carbon sequestration, Conference on Durability of Infrastructures, Mendrisio, Switzerland, 22-28 March 2025.
- [3] **B. Xi**, et al., Influence of self-healing property of Ultra-High Performance Concrete under aggressive environment, SMARTINCS'23 Conference on Self-Healing, Multifunctional and Advanced Repair Technologies in Cementitious Systems, Ghent, Belgium, 2023.

- [4] **B. Xi**, et al., Evolution of performance of self-healing in UHPC under aggressive exposure environments, 8th International Conference on Self Healing Materials (ICSHM), Milan, Italy, 2022.
- [5] **B. Xi**, et al., Investigation of self-healing property of UHPC with sustained loads under aggressive exposure environments, 14th PhD Symposium in Civil Engineering, Rome, Italy, 2022.
- [6] **B. Xi**, Y. Zhou, L. Sui, P. Li, C. Chen, B. Hu, Development of high strain hardening lightweight engineering cementitious composites, 7th Asia-Pacific Conference on FRP in Structures (APFIS 2019), Gold Coast, Australia, 2019.

INVITED LECTURES

- Invited presentation, Machine learning-accelerated discovery of green cement recipes, Global CementAI Virtual Seminar 2025, Global Cement Magazine, Online, 7 October 2025.

INTERNATIONAL MEDIA COVERAGE

- **AI-assisted low-carbon cement research (2025).** Research on *Machine learning-accelerated discovery of green cement recipes* received international media coverage following a press release by the Paul Scherrer Institute (PSI), with dissemination across Switzerland, Germany, Austria, Italy, the United Kingdom, and the United States. Selected coverage: PSI, Swiss Federal Government (admin.ch), Informationsdienst Wissenschaft (idw), Yahoo News, and other international science and technology media.